HINTS:

Conversion

60 seconds in 1 minute

60 minutes in 1 hour

1000 metres in 1 kilometre

Converting speed from km/h (kilometres per hour) to m/s (metres per second)

km/h to m/s $\div 3.6$

m/s to km/h $\times$ 3.6

Formula:

The speed of a body is the rate of chance of distance by an object.

Speed= $\frac{d}{t}$ $where\ d = distance \in metres$

t = time in seconds

The velocity of a body is the rate of change of displacement of the body.

$v = \frac{s}{t}$ *where s= displacement in metres*

t = time in seconds

Worked Example:

What is the average speed of a sprinter who runs 200 m in 15.3s?

Speed and Velocity

Name:_____

1. Conversion –find the number of: (a) Metres in 6.2 km (b) Seconds in 3.1 minutes (c) Seconds in 2.5 hours (d) Metres in 176 cm (e) Kilograms in 25 grams	2. Convert the following: (a) 100 km/h into m/s (b) 60 km/h (c) 20 km/h into m/s (d) 10 m/s into km/h
3. What is the average speed of Usain Bolt who in Berlin 2009 ran the fastest 100 m in 9.58 s in m/s and km/h?	4. A car travels 400 m in 20 s. Calculate the average speed of the car in m/s and km/h.
5. Ben rides to school on his bike from his home 8 km North East of Applecross Senior High School in 10 minutes. Calculate his average velocity.	6. A Jetstar flight travels at an average speed of 650 km/h from Perth to Hobart and take 5 h and 25 min. How far is Hobart from Perth?